NETWORK SECURITY

Week 2 – Lecture 2: Layer 2 (Ethernet) Security and Protection

Department: Cybersecurity Engineering Techniques | Stage: Fourth Year

Learning Outcomes

- Explain the role of Layer 2 and the main fields of an Ethernet frame.
- Explain how a switch learns Media Access Control (MAC) addresses and forwards frames.
- Recognize major Layer 2 attacks: MAC flooding/spoofing, (Address Resolution Protocol) ARP spoofing, Dynamic Host Configuration Protocol (DHCP) attacks, Virtual Local Area Network (VLAN) hopping, and Spanning Tree Protocol (STP) manipulation.
- Match each attack with basic protection mechanisms such as Port Security, DHCP Snooping, Dynamic ARP Inspection (DAI), Bridge Protocol Data Unit (BPDU) Guard, Root Guard, and IEEE 802.1X.

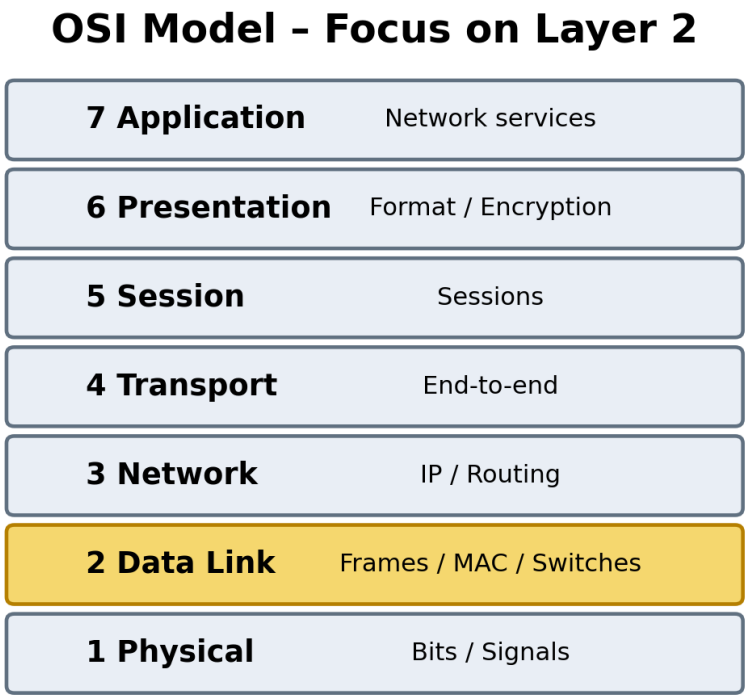


Figure 1. Open Systems Interconnection (OSI) model overview. This lecture focuses on Layer 2 (Data Link).

Key idea: Layer 2 provides local network communication using frames, MAC addresses, switches, and VLANs.

1. Ethernet and the Ethernet Frame

Ethernet is the most common wired LAN technology. At Layer 2, information is carried in units called frames. A simplified Ethernet frame contains destination and source MAC addresses, a Type/Length field, a payload, and a Frame Check Sequence (FCS).

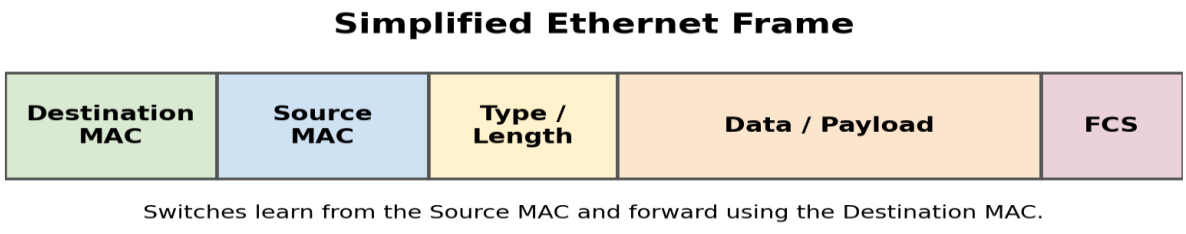


Figure 2. Simplified Ethernet frame structure.

Important Frame Fields

- Destination MAC: identifies the intended receiver.
- Source MAC: identifies the interface that sent the frame; switches use it for learning.
- Type/Length: identifies the carried protocol (for example IPv4, IPv6, or ARP).
- Payload: the actual information being transported.
- FCS: detects transmission errors; damaged frames are normally discarded.

MAC Address vs. IP Address

MAC Address	IP Address
Works mainly at Layer 2	Works at Layer 3
Used for local frame delivery	Used for communication between networks
Used by switches	Used by routers
Example: 00:25:96:FF:12:34	Example: 192.168.10.5

Key idea: A MAC address is useful for local delivery, but it is not strong proof of identity because software can change or imitate it.

2. How an Ethernet Switch Works

A switch keeps a MAC address table that maps MAC addresses to switch ports. When a frame arrives, the switch learns the source address and then looks for the destination address.

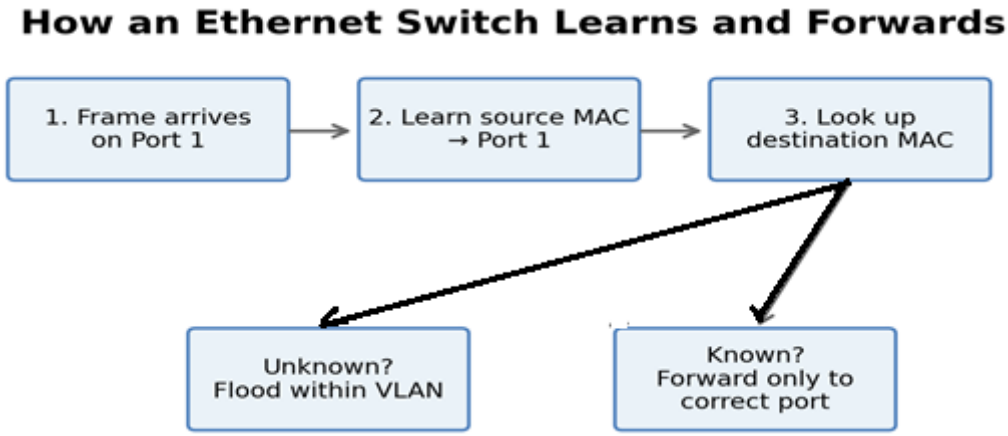


Figure 3. Switch learning and forwarding logic.

Three Key Behaviors

Situation	Switch Action
Known destination MAC	Forward the frame only to the correct port.
Unknown destination MAC	Flood the frame to other ports in the same VLAN, except the incoming port.
Broadcast destination FF:FF:FF:FF:FF:FF	Send to all ports in the same broadcast domain (except the incoming port).

Key idea: The switch learns from the source MAC address, but forwards based on the destination MAC address.

Why Layer 2 Needs Security

Many Layer 2 protocols were designed for fast local communication and may trust information received from connected devices. For example, a switch learns source MAC addresses, ARP accepts IP-to-MAC information, DHCP clients accept configuration, and STP exchanges control messages (BPDUs).

- If an attacker gains local network access, these trusted behaviors can be abused.
- Possible security effects: confidentiality, integrity, availability, and authentication.

3. MAC Flooding and MAC Spoofing

MAC Flooding

In a MAC flooding attack, an attacker sends many Ethernet frames with large numbers of false source MAC addresses. The objective is to fill or disturb the switch MAC table. Unknown unicast traffic may then be flooded to other ports, allowing unintended traffic exposure and possible performance degradation.

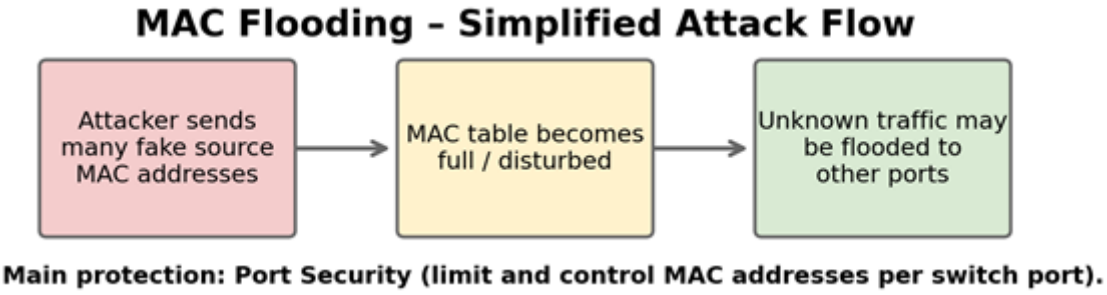


Figure 4. Simplified MAC flooding attack flow.

Attack	Main Goal / Effect	Main Protection
MAC flooding	Disturb the MAC table; may expose traffic or reduce performance	Port Security
MAC spoofing	Imitate another device by using its MAC address	802.1X, Port Security, stronger authentication

Port Security – What It Can Do

- Limit the number of MAC addresses allowed on a port.
- Use static, dynamically learned, or sticky secure MAC addresses.
- Drop unauthorized traffic or disable a port when a violation occurs.

Key idea: Do not depend on a MAC address alone for authentication. A secure network should authenticate the user or device using stronger mechanisms such as IEEE 802.1X.

4. ARP and ARP Spoofing

ARP (Address Resolution Protocol) discovers the MAC address associated with an IPv4 address on the local network. Basic ARP does not strongly authenticate every reply, so an attacker can send false IP-to-MAC information.

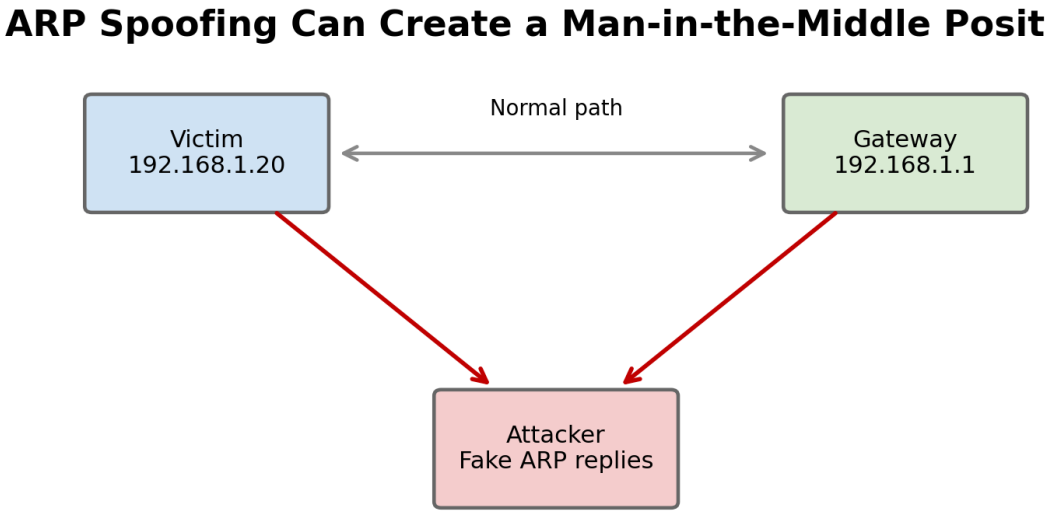


Figure 5. ARP spoofing can redirect traffic through an attacker (Man-in-the-Middle).

Example

Item	Value
Gateway IP	192.168.1.1
Real Gateway MAC	AA:AA:AA:AA:AA:01
Attacker MAC	CC:CC:CC:CC:CC:03
False ARP claim	192.168.1.1 is at CC:CC:CC:CC:CC:03

Possible Results

- Traffic interception
- Traffic modification
- Session hijacking
- Denial of service (DoS)

Protection

- Dynamic ARP Inspection (DAI)
- DHCP Snooping
- Network segmentation
- Encryption of application traffic
- Monitoring for unusual or duplicate ARP entries

Key idea: DAI validates ARP information using trusted IP-to-MAC bindings, commonly learned from the DHCP Snooping database.

Visual reference: Wikimedia Commons – "ARP Spoofing.svg" (CC BY-SA 3.0).

5. DHCP Attacks

DHCP automatically provides clients with network settings such as an IP address, subnet mask, default gateway, Domain Name System (DNS) server, and lease duration.

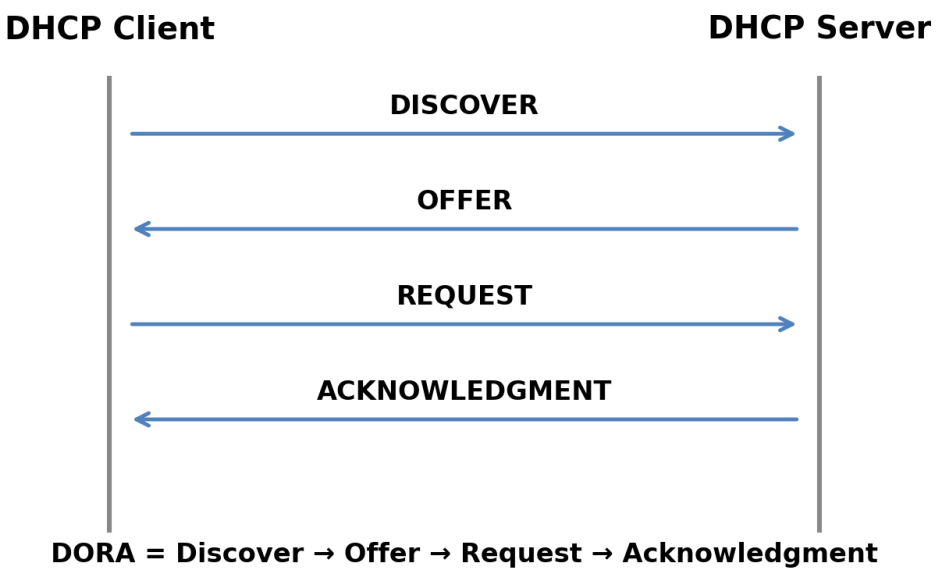


Figure 6. Normal DHCP DORA process.

Two Important DHCP Attacks

Attack	What Happens	Effect / Protection
Rogue DHCP Server	Unauthorized server gives false settings such as a fake gateway or malicious DNS server.	Traffic can be redirected. Use DHCP Snooping.
DHCP Starvation	Attacker sends many fake DHCP requests to consume the address pool.	Legitimate users may not obtain IP addresses. Use DHCP Snooping and rate limiting.

DHCP Snooping

- Classifies switch ports as trusted or untrusted.
- Blocks DHCP server messages from unauthorized untrusted ports.
- Builds a binding database containing IP, MAC, VLAN, port, and lease information.
- Supports Dynamic ARP Inspection and IP Source Guard.

Key idea: Trusted ports normally lead toward an authorized DHCP server; user-facing ports are normally untrusted.

6. VLAN Hopping and STP (Spanning Tree Protocol) Attacks

VLAN Segmentation

A VLAN logically separates a switched network into different broadcast domains. Different user groups can be placed in separate VLANs, reducing unnecessary exposure and allowing different security policies.

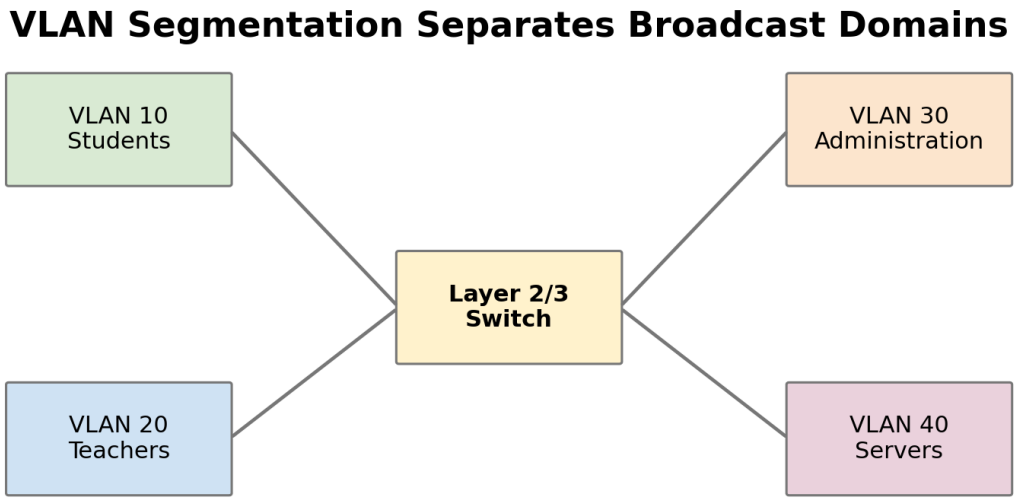


Figure 7. Example of VLAN segmentation in a university network.

VLAN Hopping

Type	Idea	Basic Protection
Switch spoofing	An attacker tries to appear as a switch and negotiate a trunk.	Configure user ports as access ports; disable automatic trunk negotiation.
Double tagging	A frame contains two VLAN tags and may cross VLAN boundaries under vulnerable conditions.	Use a dedicated unused native VLAN; avoid user devices on the native VLAN.

STP Manipulation

STP prevents switching loops and selects a root bridge. An unauthorized switch may send superior BPDUs to try to become the root bridge, change topology, redirect traffic, or cause instability.

- BPDU Guard: commonly used on host-facing access ports; unexpected BPDUs can disable the port.
- Root Guard: prevents an unexpected switch from becoming the STP root through a protected port.

Key idea: BPDU Guard protects edge ports; Root Guard protects the expected root-bridge position.

7. Main Layer 2 Protection Mechanisms

Control	Main Purpose
Port Security	Limits which MAC addresses can use a switch port.
Disable Unused Ports	Reduces unauthorized physical network access.
VLAN Segmentation	Separates users/services into security zones.
Dynamic Host Configuration Protocol (DHCP) Snooping	Blocks unauthorized DHCP server traffic and builds trusted bindings.
Dynamic ARP Inspection (DAI)	Validates ARP messages against trusted binding information.
IP Source Guard	Filters source IP/MAC information on untrusted interfaces.
Bridge Protocol Data Unit (BPDU) Guard	Stops unexpected STP participation on user-facing ports.
Root Guard	Protects the intended STP root position.
IEEE 802.1X	Authenticates users/devices before normal network access is allowed.

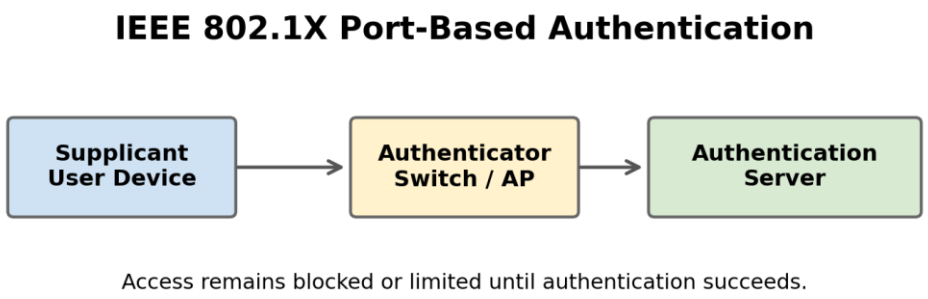


Figure 8. Simplified IEEE 802.1X authentication flow.

Until authentication succeeds, normal network access can be blocked or limited.

IEEE 802.1X – Three Main Components

Component	Role
Supplicant	The user device requesting network access.
Authenticator	The switch or access point controlling the port.
Authentication Server	The server that verifies credentials.

Key idea: Layer 2 security works best when several controls are combined. No single control can stop every attack.

8. Defense in Depth + University Case Study

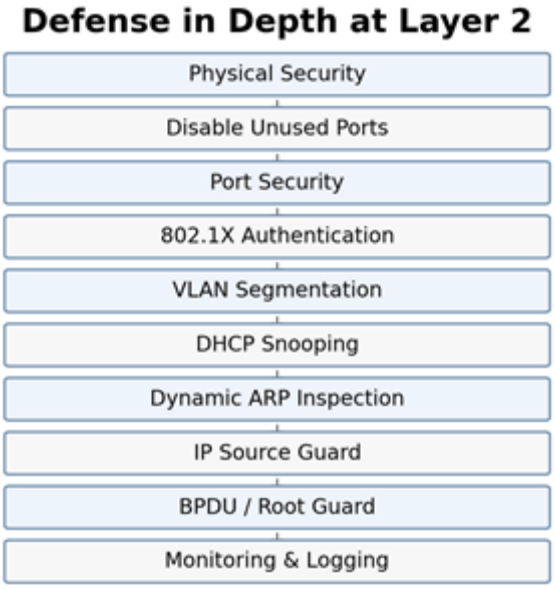


Figure 8. Example of Defense in Depth at Layer 2.

University Network Case Study

Current Problem	Recommended Protection
All users are in one VLAN	Create separate VLANs for students, teachers, administration, servers, and guests.
Unused switch ports are active	Administratively shut down unused ports and place them in an unused VLAN.
Unknown devices can connect	Use IEEE 802.1X.
Too many MAC addresses appear on one port	Enable Port Security.
Rogue DHCP server is possible	Enable DHCP Snooping.
ARP spoofing is possible	Enable Dynamic ARP Inspection.
Students may connect unauthorized switches	Use BPDU Guard.
Unexpected switch may become root	Use Root Guard where appropriate.

Key idea: Defense in Depth means that if one protection fails, another security layer may still reduce the attack or limit its impact.

9. Attack–Protection Summary

Layer 2 Attack	Main Objective	Suitable Protection
MAC flooding	Fill or disturb the MAC table	Port Security
MAC spoofing	Imitate another device	802.1X, Port Security
(Address Resolution Protocol)ARP spoofing	Redirect or intercept traffic	Dynamic ARP Inspection (DAI), DHCP Snooping, encryption
Rogue DHCP	Give clients false configuration	DHCP Snooping
Bridge Protocol Data Unit (DHCP) starvation	Exhaust available IP addresses	DHCP Snooping, rate limiting
VLAN hopping	Reach another VLAN	Fixed access mode, disabled negotiation, secure native VLAN
STP manipulation	Change switching topology	BPDU Guard and Root Guard
Unauthorized connection	Gain network access	Disable unused ports, 802.1X

Remember These Principles

- Do not trust a device only because it is connected to the local network.
- A MAC address is not strong proof of identity.
- Separate users and services using VLANs.
- Disable unused ports and unnecessary trunk negotiation.
- Authenticate users/devices before providing access.
- Protect ARP, DHCP, VLAN, and STP operations and monitor switch logs.

Quick Review Questions

1. How does a switch learn a MAC address?
2. What is the difference between MAC flooding and MAC spoofing?
3. How can ARP spoofing create a Man-in-the-Middle position?
4. What is the DORA process in DHCP?
5. What is the difference between BPDU Guard and Root Guard?
6. Which controls protect against rogue DHCP and ARP spoofing?

References and Visual References

- W. Stallings, Network Security Essentials: Applications and Standards, Pearson.
- Cisco Networking Academy, Network Security and Switching Security Materials.
- IEEE 802.3 (Ethernet) and IEEE 802.1 (bridging, VLANs, and port-based access control).
- Wikimedia Commons: “OSI Model v1.svg” (CC0) and “ARP Spoofing.svg” (CC BY-SA 3.0) were consulted as visual references.